Prob.4. (2&T, Ch.7.1.3) (?5) $u \in C^0$ or C^2 ?

Be sure to show $u \in C^2(\Omega)$. Recall that 2&T uses $\log(\cdot)$ for $\ln(\cdot)$.

(a) Show that $u = \ln(x^2+y^2)$ is harmonic on $\Omega = \mathbb{R}^2 \setminus (0,0)$.

sol.n

Based on the props of harmonic f-ns in Prob.1, let's show

① $u \in C^2(\Omega)$ & (Smoothness)

② $\Delta u = 0$ (satisfies the Laplace eq-n);

(1) $\boxed{\Omega = \mathbb{R}^2 \setminus \{\underline{0}\}}$

• for ①, recall

$C^0(\Omega) = \{f(x) : f(x) \text{ is cont-s over } \Omega\}$

$C^1(\Omega) = \{f(x) : f(x) \in C^0(\Omega) \text{ \& } \frac{\partial f}{\partial x_i} \in C^0(\Omega)\}$ &

(2) $\boxed{C^2(\Omega) = \{f(x) : f(x) \in C^0(\Omega), \frac{\partial f}{\partial x_i} \in C^0(\Omega), \frac{\partial^2 f}{\partial x_i \partial x_j} \in C^0(\Omega)\}}$

$C^\infty(\Omega) = \{f(x) : f(x) \in C^0(\Omega)$ & all (from 1st order up to ∞ order) partial derivatives are $\in C^0(\Omega)\}$.

Note that first,

(3) $\{u(x,y) = \ln(x^2+y^2) \in C^0(\mathbb{R}^2 \setminus \{\underline{0}\})$

✓

Now 1st order partials,

(4)
$$\begin{cases} u_x = \frac{1}{x^2+y^2} \cdot (2x) = \frac{2x}{x^2+y^2} \in C^0(\mathbb{R}^2 \setminus \{\underline{0}\}) \text{ \&} \\ u_y = \frac{1}{x^2+y^2}(2y) = \frac{2y}{x^2+y^2} \in C^0(\mathbb{R}^2 \setminus \{\underline{0}\}) \quad (\text{as denom.} \neq 0) \end{cases}$$

And finally 2nd order partials;

(5)
$$\begin{cases} u_{xx} = \frac{2y^2-2x^2}{(x^2+y^2)^2} \in C^0(\mathbb{R}^2 \setminus \{\underline{0}\}) \quad (\text{as denom} \neq 0) \\ u_{xy} = \frac{-4xy}{(x^2+y^2)^2} \in C^0(\mathbb{R}^2 \setminus \{\underline{0}\}) \quad (\text{denom} \neq 0) \text{ \&} \\ u_{yy} = \frac{2x^2-2y^2}{(x^2+y^2)^2} \in C^0(\mathbb{R}^2 \setminus \{\underline{0}\}) \end{cases}$$

(3), (4), (5) meet all requirements of $u \in C^2(\mathbb{R}^2 \setminus \{\underline{0}\})$.

- For ② ($\Delta u = 0$)

(6) $$\Delta u = u_{xx} + u_{yy} \overset{\text{using (5)}}{=} \frac{2y^2-2x^2}{(x^2+y^2)^2} + \frac{2x^2-2y^2}{(x^2+y^2)^2} = \frac{0}{(x^2+y^2)^2} \equiv 0 \checkmark$$

u also satisfies the Laplace eq-n, hence u is harmonic. ∎

(b) Show that $u(x,y) = (x^2+y^2+z^2)^{-\frac{1}{2}}$ is harmonic over $\Omega = \mathbb{R}^3 \setminus \{0,0,0\}$.

Sol-n.

(Recall the Property 1 of harmonic f-ns: if $u(x,y)$ is cont-s & satisfies the Laplace eq-n, then $u(x,y) \in C^\infty(\Omega)$. So in particular, $u \in C^2(\Omega)$, & thus it is sufficient to show that $u(x,y)$ is cont-s & satisfies the Laplace eq-n.)

(i) $$u = \frac{1}{\sqrt{x^2+y^2+z^2}} \in C^2(\mathbb{R} \setminus \{\underline{0}\}) \quad \&$$

(ii) $$u_x = \frac{-x}{\sqrt{(x^2+y^2+z^2)^3}}, \quad u_y = \frac{-y}{\sqrt{(x^2+y^2+z^2)^3}}, \quad u_z = \frac{-z}{\sqrt{(x^2+y^2+z^2)^3}} \Rightarrow$$

$$u_{xx} = \frac{2x^2-y^2-z^2}{(x^2+y^2+z^2)^{\frac{5}{2}}}, \quad u_{yy} = \frac{2y^2-x^2-z^2}{(x^2+y^2+z^2)^{\frac{5}{2}}}, \quad u_{zz} = \frac{2z^2-x^2-y^2}{(x^2+y^2+z^2)^{\frac{5}{2}}}, \text{ so}$$

$$\Delta u = u_{xx} + u_{yy} + u_{zz} = \frac{(2x^2-y^2-z^2)+(2y^2-x^2-z^2)+(2z^2-x^2-y^2)}{(x^2+y^2+z^2)^{\frac{5}{2}}} = \frac{0}{\boxed{\cdots}} = 0$$

Hence u satisfies 2 conditions to be a harmonic f-n. ∎